Translational Dynamics

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The primary objective of classical physics is to determine the future state of a physical system or object. In kinematics, we learn to determine an object's future position, velocity, and other related variables given its acceleration. Dynamics, on the other hand, deals with the motion of objects under the influence of various forces. In this section, we will focus on Newtonian mechanics. Please find the latest version of the note here.

1 Newton's Laws

The foundation of Newtonian mechanics is set by Newton's three laws of motion. Let us first review these laws:

1. **First Law:** In an inertial reference frame, the velocity of an object remains constant unless it is acted upon by a non-zero net external force $\left(\sum \vec{F} \neq 0\right)$.
2. **Second Law:** The net force applied to an object is equal to the rate of change of its momentum, $\vec{p}$, with respect to time. That is, $\sum \vec{F} = \frac{d\vec{p}}{dt}$
3. **Third Law:** For every applied force, there is an equal and opposite force.

These laws form the fundamental basis of Newtonian mechanics. We will now discuss each law in detail.

First Law

This law essentially defines the concept of an inertial reference frame. You can interpret the law as follows: a reference frame in which an object experiences zero acceleration when the net external force acting on it is zero–is called an inertial reference frame. Newton's second law is strictly valid only in these inertial frames. While Newton's laws can be applied in non-inertial reference frames, doing so requires certain modifications to the equations. We will discuss this in greater detail in another note.
For now, consider an example where you are sitting in a car that is accelerating forward. There are no real horizontal force acting on you, but you still feel a backward push and tend to move backward relative to the car. Since you experience an acceleration despite the net real force being zero, the accelerating car is a non-inertial frame. For most practical purposes, we approximate the Earth as an inertial frame. Furthermore, any reference frame moving at a constant velocity relative to an inertial frame is also an inertial frame.

Second Law

According to Newton's second law, in an inertial reference frame:

$$\sum \vec{F} = \frac{d\vec{p}}{dt}$$

where $\sum \vec{F}$ is the net external force acting on the object and $\vec{p}$ is its momentum. We know that, $\vec{p} = m\vec{v}$, where m is the mass of the object and $\vec{v}$ is its velocity. If m remains constant, this can be rewritten as:

$$\sum \vec{F} = m\vec{a}$$

Note that, at this stage, we are only considering point masses. We will learn how to apply Newton's second law to extended bodies in a later section. Furthermore, because Newton's second law is a vector equation, it can be resolved into its components to three scalar equations (one for each spatial axis).

Third Law

If Object 1 exerts a force $\vec{F}_{21}$[1] on Object 2, Newton's third law states that Object 2 will also exert a force on Object 1. This force is given by $\vec{F}_{12} = -\vec{F}_{21}$.

Note that, this action-reaction pair of forces never acts on the same object. From this, we can deduce that if we have a system of two isolated particles interacting only with each other (with no external forces applied), then:

$$\frac{d\vec{p}_{total}}{dt} = \sum \vec{F}$$

$$\Rightarrow \frac{d\vec{p}_{total}}{dt} = \vec{F}_{12} + \vec{F}_{21} = 0$$

Therefore, the total momentum of the system remains constant. Because Newton's third law is directly linked to the conservation of momentum, it is generally considered universally true within classical mechanics.

2 Center of mass

Before moving on to problem-solving, we need to discuss the concept of center of mass. Until now, we have been analyzing single point masses. If a system contains multiple objects, we could technically apply Newton's second law to each object individually. However, tracking many interacting particles makes the mathematics highly complex. Instead, we can imagine the entire collection of objects as an unified system with a total mass M (the sum of all individual masses) located at a specific coordinate $\vec{R}$. This point is called the Center of Mass (CM) of the system. The advantage of this approach is that we no longer need to account for the internal forces between the particles; according to Newton's third law, these internal forces perfectly cancel each other out in pairs. To determine the acceleration of the center of mass, we only need to consider the net external force acting on the system. For rigid bodies, the center of mass is even more convenient; since the relative distances between the particles in a rigid body do not change, the translational motion of the entire object can be completely described by tracking its center of mass. Let us now derive the equation for this point. Suppose our system consists of n point masses, having masses $m_1, m_2, \cdots, m_n$ and position vectors $\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_n$, respectively. According to Newton's second law,

$$\vec{F} = \sum_{i=1}^{n} m_i \ddot{\vec{r}}_i$$

[1] Pay attention to the notation here: the first subscript denotes the object experiencing the force, and the second subscript denotes the object exerting it. Like all notations, this is merely a convention; you are free to use a different convention, provided you remain consistent.

Furthermore, from the definition of the center of mass, we can write $\vec{F} = M\ddot{\vec{R}}_{cm}$, where $M = \sum m_i$. Equating the two yields:

$$M\ddot{\vec{R}}_{cm} = \sum_{i=1}^{n} m_i \ddot{\vec{r}}_i$$

Integrating this equation twice with respect to time[2], we obtain:

$$\vec{R} = \frac{1}{M}\sum_{i=1}^{n} m_i \vec{r}_i$$

This is the fundamental equation for the center of mass. When dealing with systems containing two or more interacting objects, shifting your coordinate system to the center of mass reference frame often simplifies the problem significantly.

To find the center of mass of a solid, continuous object, we can first picture the object as a continuous distribution of an infinite number of infinitesimal mass elements, dm. We then multiply each infinitesimal mass by its position vector, sum over all mass of the object, and divide by the total mass:

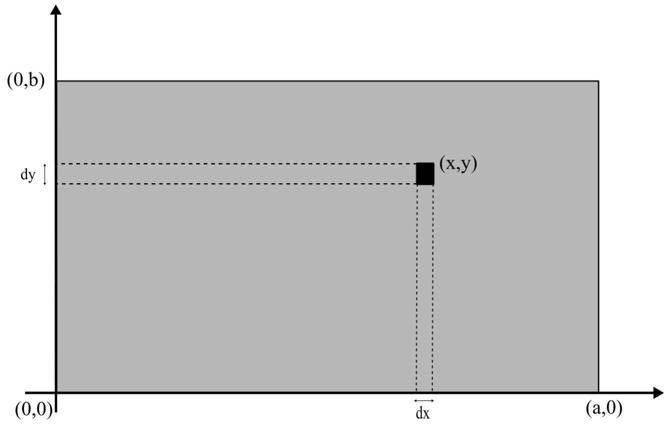
$$\vec{R} = \frac{1}{M}\int_V \vec{r}dm$$

Additionally, if an extended body is composed of several simpler parts, the center of mass of the entire composite structure can be found by treating each individual part as a point mass located at its own respective center of mass. Try proving this mathematical shortcut yourself as an exercise!

As mention earlier, for rigid bodies, the center of mass offers a unique advantage because the relative positions of all other points in the body remain fixed with respect to it, changing only through rotation. The general motion of any rigid body can be completely decomposed into the transnational motion of its center of mass and its rotation about that center of mass. If there is no rotation, the displacement, velocity, and acceleration of the center of mass alone are sufficient to define the motion of the object. Simply put, we can mathematically replace the entire extended object with a single point mass located at its center of mass! Rigorously calculating the center of mass for completely arbitrary shapes can be mathematically complex. Fortunately, in most practical physics problems, you won't need to perform an explicit integration. You can usually deduce the location of the center of mass simply by looking at the object's geometry and utilizing symmetry. Nevertheless, we will demonstrate the general calculation procedure by calculating the center of mass of a uniform rectangle next. However, even if you do not fully grasp the example, do not worry. You can easily skip it and proceed to the next section!

Example: Calculate the center of mass of a rectangle with mass M, length a, and width b.

[2] Here, we can determine the constants of integration by considering the $n = 1$ case. If there is only one particle in the system, then $\vec{R}$ and $\vec{r}_1$ must be identical.



Let us set the origin at the bottom-left corner of the rectangle and imagine the rectangle as a collection of infinitesimally small mass elements with length dx and width dy. Now,

$$dm = \frac{M}{ab}dxdy$$

$$\begin{aligned}
\therefore \vec{R} &= \frac{1}{M}\int_0^a \int_0^b (x\hat{i} + y\hat{j})\frac{M}{ab}dxdy \\
&= \frac{1}{M}\left[\int_0^a \int_0^b \left(x\frac{M}{ab}dxdy\right)\hat{i} + \int_0^a \int_0^b \left(y\frac{M}{ab}dxdy\right)\hat{j}\right] \\
&= \frac{1}{M}\left[\frac{M}{ab}\left[\frac{x^2}{2}\right]_0^a [y]_0^b\,\hat{i} + \frac{M}{ab}\left[\frac{y^2}{2}\right]_0^b [a]_0^a\,\hat{j}\right] \\
&= \frac{1}{ab}\left(\frac{a^2}{2}\cdot b\hat{i} + \frac{b^2}{2}\cdot a\hat{j}\right) \\
&= \frac{1}{2}(a\hat{i} + b\hat{j})
\end{aligned}$$

Notice that the center of mass we just obtained is exactly the geometric center of the rectangle! As a general rule, for any object with a uniform density, its center of mass will always coincide with its geometric center. We will not deal with any problems involving center of mass in this chapter, but you would see many applications in other chapters, including the chapter on Conservation laws and Central forces.

3 Different types of forces

Before working with forces, it is necessary to understand the different types of forces.

Tension

The force generated within objects like ropes or rods when they are pulled or pushed, acting in the opposite direction of the pull/pull, is tension. Every small segment of a rope experiences tension in both directions, except at the two ends. As a result, the net force on any part other than the ends is zero. Any other object attached to the ends experiences this tension force. In the case of a massless rope on a frictionless medium, the tension is uniform throughout the entire rope. If the tension were not equal at all points, there would be a non-zero net force at some point. If a non-zero net force acts on a massless rope, its acceleration would be $\frac{F}{0}$, which is infinite. Since infinite acceleration is physically impossible, the tension must remain constant. Note that this does not mean the acceleration of the rope will always be zero. If the tension is constant, the net force on every segment is zero,

and the acceleration takes the indeterminate form of $\frac{0}{0}$. Therefore, the rope can move with any real, finite acceleration. If the rope has its own mass and is accelerating, a net force is required to accelerate each infinitesimal segment of the rope. In that case, the magnitude of the tension will vary at different points along the rope.
There is no general equation for calculating the tension in a rope. Tension exists in a system to satisfy a specific condition or constraint that we impose. Usually, this constraint is that the length of the rope must remain unchanged. However, if we specify that the rope changes its length in a particular way, we can also determine the tension using that constraint. Let us try to understand this better with an example. Suppose we connect two blocks with an inextensible rope. We then begin pulling the first block with a certain acceleration. Since the length of the rope cannot change, the first block must pull the second block along with it as it moves forward! Now, which force provides the acceleration for the second block? It is the tension from the attached rope! The magnitude of this tension will be exactly what is needed to ensure both blocks have the same acceleration. We can determine the value of the tension using this very condition. Do not worry if this is not completely clear yet; we will see more examples later.

Normal Force

Any surface can exert two types of forces on another object or surface. These can be broadly categorized as contact forces. The first acts parallel to the surface, which we call the friction force. The second acts perpendicular to the surface, which we call the Normal force. The term "Normal" is used because its mathematical meaning is "perpendicular." This name implies that the force acts perpendicularly. When a force is applied to a surface, the reactive force felt perpendicular to the surface is the normal force. In the case of a curved surface, the normal force acts perpendicularly to the tangent at the point of contact. Note that the normal force always pushes outward from the surface, and its magnitude is never negative. If the normal force on a surface becomes $N = 0$, it means the object has lost contact with the surface.
Like tension, the normal force arises in our system due to constraints. Usually, this constraint is that there should be no acceleration perpendicular to the surface (i.e., the surface/table will not break!). When you sit on a chair, a total downward force of Mg acts on you. Therefore, you should be breaking the chair and falling downward with an acceleration of g! However, we assume the chair is strong enough and will not break. To fulfill this condition, your downward acceleration on the chair must be zero, and the perpendicular force created to ensure this is what we call the Normal force. We will determine its value by setting up equations based on these constraints, and naturally, its value will vary under different conditions (the equations for normal force will be different in an elevator, on an inclined plane, or on a flat surface). This concept will become much clearer after the 'Equation of constraint' section.

Friction force

The force exerted by a surface parallel to itself (or parallel to the tangent at the point of contact, for a curved surface) is called the friction force. The origin of this force is that no surface is perfectly smooth, and rough surfaces oppose the relative motion of objects across them. This force always acts in the opposite direction of the relative motion. In general frictional forces are very difficult to model. We will use the simplest of all models for friction. In this model, all friction can be mathematically divided into two categories: kinetic friction and static friction. In kinetic friction, the force opposing the relative motion is:

$$F_{fric,kin} = \mu_k N$$

where N is the normal force and μ_k is the coefficient of kinetic friction. Its value depends on the materials of the surfaces and cannot be determined without experiments. The magnitude of the kinetic friction force is constant for a given μ_k and N. For static friction:

$$F_{fric,stat} \leq \mu_s N$$

Here, μ_s is the coefficient of static friction. When there is no relative motion between the two surfaces, static friction is at play. Notice that the value of static friction is not equal to $\mu_s N$; it can be less than that. When an attempt is made to move an object relative to another, this friction provides resistance. If the applied force is less than $\mu_s N$, the magnitude of the friction force is exactly equal to the applied force. When the applied force reaches $\mu_s N$, the friction force also reaches its maximum limit. If the applied force exceeds $\mu_s N$, relative motion begins between the two surfaces, and static friction is replaced by kinetic friction. Whenever there is relative motion between two surfaces, kinetic friction acts. For the same pair of materials, the coefficient of kinetic friction is usually lower than the coefficient of static friction.

Gravitational Force

The attractive force that acts between two objects purely due to their masses is the gravitational force. According to Newton's law of universal gravitation, the magnitude of this force is:

$$\vec{F}_{grav} = -\frac{GMm}{r^2}\hat{r}$$

Here, G is the gravitational constant, M and m are the masses of the two objects, and r is the distance between them.

Spring Force

The spring force is the opposing force exerted by a spring when it is compressed or stretched. Mathematically:

$$\vec{F}_{spring} = -kx\hat{x}$$

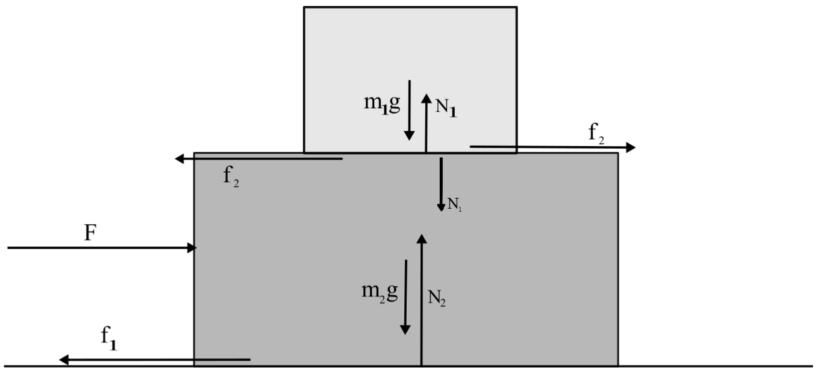
where x is the compression or extension of the spring from its equilibrium position, and $\hat{x}$ is the unit vector in the direction of the displacement from equilibrium.

Additionally, there are other non-mechanical forces, such as the Coulomb force and the magnetic force. We will not be discussing these types of forces here. Remember, there are no direct formulas for determining tension and normal force. They usually act as variables, and their values are found by setting up and solving a sufficient number of equations based on system constraints. The other forces have their own specific formulas.

4 Free Body Force Diagram

When solving problems in Newtonian mechanics, you will typically find multiple forces acting on the same object. In such cases, it is very common to lose track of which force is acting where. To avoid this problem, we will use a Free Body Diagram. This is essentially a diagram where you draw an object and all the forces acting upon it as vectors. Unknown forces should also be represented in the diagram using variables. If you do not know the magnitude of a force, you will treat it as a variable; however, do not blindly introduce new variables without first checking if the value can be determined in some way! For example, sometimes you will spot an action-reaction force pair from Newton's Third Law. Draw them as vectors with the same magnitude but acting in opposite directions. Do not introduce unnecessarily many variables.

Imagine a block of mass m_1 placed on top of a block of mass m_2. The bottom block is being pushed with a force F. The coefficient of friction for all surfaces is μ. Observe the force diagram for this problem:



Here, the forces acting on each object have been placed with their correct directions. Another point should be mentioned here: an attempt has been made to draw each force at its point of application. The point of application is essentially the exact point where the force acts. The normal force and weight will act along the same vertical line, but for the convenience of drawing, they are shown side by side. Also notice that the friction force acting between the two blocks is a Newton's Third Law force pair. Therefore, both have been denoted by the value f_2. [3] Once the force diagram is drawn correctly, your only task is to take the components of the forces along different axes and apply $F = ma$ along those axes.
With what we have learned so far, we can solve many types of problems. Let us outline the methods for solving problems:

In a very small number of problems, the force is given as a function of something, and the acceleration needs to be determined as a function of a specific quantity. These problems are purely mathematical. Simply writing $\vec{F} = m\vec{a}$ and solving it usually suffices. Solving these requires some mathematical skills, which we will discuss in the next section. In almost all other types of problems, a system of one or more objects is usually given, where the objects exert forces on each other in various ways. The general approach for these types of problems will be as follows:

- First, you must draw a force diagram for the problem. You need to draw all known and unknown possible forces acting on it in this diagram. If you are unsure whether a force is actually acting, draw it anyway, but keep in mind later that its value might be zero.

- Next, you need to select a system or systems where you will apply Newton's second law. Your system may contain just one or multiple objects. Generally, if you do not need to know the internal forces between multiple objects, it is convenient to take those multiple objects together as a single system. If you need to know the internal forces, you must form equations by treating each object as a separate system. Note that even if you don't need the internal forces, you can still solve the problem by taking each object as a separate system. However, taking a larger system makes the solution simpler and shorter.

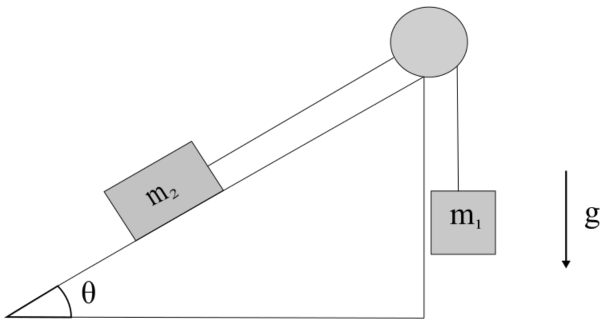
[3] The easiest way to identify such force pairs is to imagine a larger system. If you considered both blocks together as a single system, you would not need to write the friction force between them, as it would be an internal force and thus cancel out. Hence, they form an action-reaction force pair according to Newton's Third Law.

- Once the system or systems are selected, you now have to apply Newton's second law along each axis of every system. Sometimes there is no motion along certain axes, or we are not concerned with the motion along that axis; in such cases, we write Newton's second law along the axis perpendicular to it.

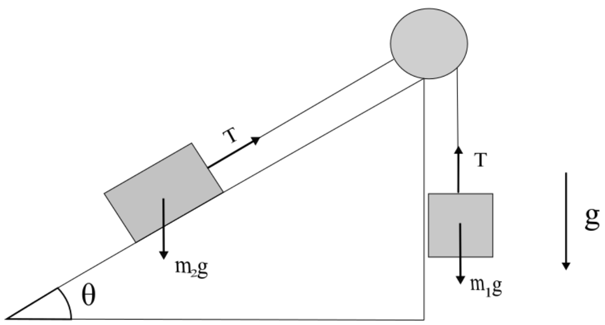
- In most problems, there is a specific relationship between the accelerations of multiple objects. These types of relationships must be written down in the form of equations. These are called equations of constraint, and we will discuss this later.

If you have followed each step correctly, you should have as many equations as you have unknown variables. Now, solve the equations using any method you prefer to determine the required quantities!

Example: As shown in the figure below, a block of mass m_2 is on a fixed, frictionless inclined plane at an angle θ. This block is connected to a mass m_1 via a massless, inextensible rope passing over a massless pulley. Determine the acceleration of both blocks and the tension in the rope.



Solution: First, let us draw a force diagram.



It is not always a good strategy to start solving a problem immediately upon seeing it. First, look at the problem and try to understand what is physically happening. With a little thought, you will realize that since the rope is inextensible, the acceleration at both ends of the rope must be the same. Let's assume the magnitude of this acceleration is a. Now, to determine the value of this acceleration a, we are not concerned with any internal forces. Therefore, we can consider the two blocks and the rope together as our total system. The system only has acceleration along the rope, so we will apply $\vec{F} = m\vec{a}$ along the rope[4]. If we

[4]Here, you might wonder whether $F = ma$ can be applied along the rope since it bends in the middle. The

consider the downward motion of m_1 as positive, then the total force on the system along the rope is,

$$\sum F = m_1 g - m_2 g \sin\theta$$

According to Newton's second law,

$$\sum F = (m_1 + m_2)a$$

$$\Rightarrow a = \frac{\sum F}{m_1 + m_2} = \frac{m_1 g - m_2 g \sin\theta}{m_1 + m_2}$$

Therefore, the acceleration of block 1 is a, vertically downwards; and the acceleration of block 2 is also a, but upwards along the plane. Note that if we get a negative value for a from the equation, it will mean that the acceleration of m_2 is downwards along the plane and the acceleration of m_1 is upwards.

Now, since the tension of the rope is an internal force within this large system, we can no longer use the previous system to determine it. Since the tension is the same throughout the rope, we can determine it by treating any single block as the system. Treating block 1 as the system and applying Newton's second law vertically,

$$m_1 g - T = m_1 a$$

Substituting the previously solved value of a,

$$\begin{aligned} T = m_1(g - a) &= m_1 \frac{m_1 g + m_2 g - m_1 g + m_2 g \sin\theta}{m_1 + m_2} \\ &= m_1 m_2 g \frac{1 + \sin\theta}{m_1 + m_2} \end{aligned}$$

5 Newton's Laws in Polar Coordinates

In the Cartesian coordinate system, Newton's second law takes a very simple form:

$$F_x = ma_x \qquad ; \qquad F_y = ma_y \qquad ; \qquad F_z = ma_z$$

However, polar coordinates are sometimes much more advantageous than Cartesian coordinates. In the kinematics chapter, we saw that acceleration in polar coordinates is given by:

$$\vec{a} = (\ddot{r} - \dot{\theta}^2 r)\hat{r} + (r\ddot{\theta} + 2\dot{r}\dot{\theta})\hat{\theta}$$

Therefore, writing the vector form of Newton's second law ($\vec{F} = m\vec{a}$) in two-dimensional polar coordinates yields:

$$F_r = m(\ddot{r} - \dot{\theta}^2 r) \qquad \text{and} \qquad F_\theta = m(r\ddot{\theta} + 2\dot{r}\dot{\theta})$$

This form will be incredibly useful when applying Newton's second law to objects moving along circles, spirals, or any other paths that are easily expressed in polar coordinates. It will be especially useful when we discuss central forces later on.
Notice that, unlike in Cartesian coordinates, we cannot simply state that $F_r = m\ddot{r}$; we must

answer is yes, it can be; notice that the pulley only exerts a force perpendicular to the rope. If there were friction between the pulley and the rope, that would also have to be taken into account when calculating the net force along the rope.

add an extra term, $-m\dot{\theta}^2 r = -\frac{mv^2}{r}$ (since in circular motion, $\dot{\theta} r = v$). Once again, the reason for this additional term is that even when r is constant, the direction of the object's velocity is continuously changing, which requires an additional acceleration of v^2/r directed toward the center. To produce this required acceleration, we need a centrally-directed force of mv^2/r. This is (unfortunately) called the centripetal force. Note that this is not actually a distinct force like the others. It merely represents the minimum amount of centrally-directed force required to keep the object moving in a circular path. For example, consider a satellite orbiting the Earth in a circular path. To keep it in this circular orbit, a centripetal acceleration of v^2/r, or equivalently a force of mv^2/r, is required – this itself is not a physical force acting towards the center! In reality, this required centripetal force is provided by some other real force, such as gravity or the tension in a rope, etc. If the total real force directed toward the center is greater than $m\dot{\theta}^2 r$, the equation above shows that we will have $\ddot{r} < 0$. If we wish, we can rewrite the equation for the $\hat{r}$ direction as:

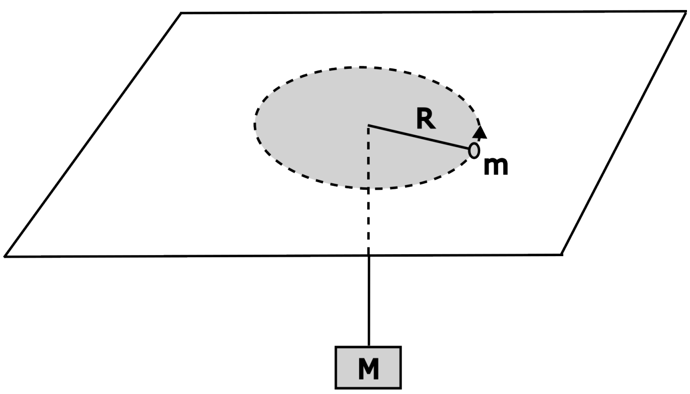
$$m\ddot{r} = F_r + m\dot{\theta}^2 r$$

That is, if we insist on writing $F = m\ddot{r}$ exactly as we do in Cartesian coordinates, we must add a fictitious, centrifugal (directed outward from the center) force $m\dot{\theta}^2 r$ to F, sum of the real forces. We call this fictitious force the centrifugal force. This centrifugal force arises only when we attempt to write Newton's laws from the perspective of a rotating reference frame. Since a rotating reference frame is non-inertial, we must make certain modifications to apply Newton's laws there, and the introduction of this fictitious force is a direct consequence of that. We will explore this in much greater detail in the chapter on Newton's laws in non-inertial reference frames. The force equation in the $\hat{\theta}$ direction can be analyzed in a similar manner, but we will reserve that discussion entirely for the non-inertial frames chapter. For now, by multiplying the angular equation by r, we can write:

$$rF_\theta = mr^2\ddot{\theta} + 2mr\dot{r}\dot{\theta} = \frac{d}{dt}(mr^2\dot{\theta})$$

Here, the term inside the parenthesis is the angular momentum of the object. We will also save the detailed discussion of rotation for a later chapter.

Example: A point particle of mass m is placed on a horizontal, frictionless table. It is tied to a block of mass M and hung below the table through a small hole on the table. Initially, a length R of the rope is on the table, and the point particle has a tangential speed v_0 along the table. Initially, the particle is traveling in a circle of radius R. The particle is then given a very small radial nudge. Is the resulting motion simple harmonic? If yes, what is the angular frequency of the oscillation?



Solution: The only force on the particle along the table is the tension force of the rope. We can use usual Cartesian coordinate to write the force equation on M:

$$Mg - T = M\ddot{y}$$

For the particle on the table, it is more convenient to use polar coordinates, with center of coordinate at the hole on the table. Force equation of the particle:

$$-T = m(\ddot{r} - \dot{\theta}^2 r) \qquad \text{and} \qquad 0 = \frac{d}{dt}(mr^2\dot{\theta})$$

Second equation gives us a conserved quantity $L = mr^2\dot{\theta} = mRv_0$, where we have used the fact that initially $r = R$ and $\dot{\theta}r = v_0$. We can use this to eliminate $\dot{\theta}$ from the radial equation:

$$\ddot{r} - \frac{v_0^2 R^2}{r^3} + \frac{T}{m} = 0$$

We cannot yet use the force equation for M to eliminate T, as it would also introduce another new variable y in the above equation. However, we can note that since the length of the rope $y + r$ has to be constant, $\ddot{y} + \ddot{r} = 0$ (more on this in the following section). Now we can readily eliminate T:

$$\left(1 + \frac{M}{m}\right)\ddot{r} - \frac{v_0^2 R^2}{r^3} = -\frac{Mg}{m}$$

The next part uses some theory that we will learn in Oscillation notes. If you are not familiar with it, feel free to just skim through this part!
As the particle is in a circle initially, we have $\ddot{r}(0) = 0$, and $r(0) = R$. After a bit of rearranging, this tells us that for the particle to move in a circle, we must have

$$Mg = \frac{mv_0^2}{R}$$

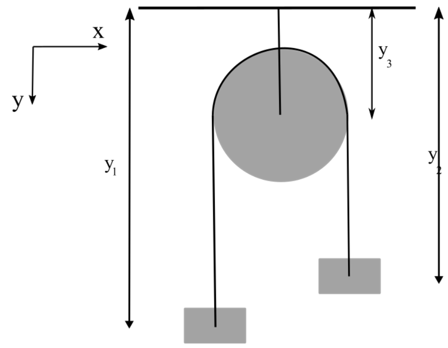
This part could've been found easily by noting that for circular orbits, the bottom mass is stationary and thus $T = Mg$, and tension force is only providing the centripetal force to the point particle. Nevertheless, consider small displacement $\eta \ll R$ from this circular orbit such that $r(t) = R + \eta(t)$. We can write:

$$\begin{aligned}
\left(1 + \frac{M}{m}\right)\ddot{\eta} - \frac{v_0^2 R^2}{R^3(1+\eta/R)^3} &= -\frac{Mg}{m} \\
(m+M)\ddot{\eta} - \frac{mv_0^2}{R}\left(1 - \frac{3\eta}{R}\right) &= -Mg \\
(m+M)\ddot{\eta} - Mg\left(1 - \frac{3\eta}{R}\right) &= -Mg \\
(m+M)\ddot{\eta} + \frac{3Mg}{R}\eta &= 0 \\
\ddot{\eta} + \frac{3Mg}{(M+m)R}\eta &= 0
\end{aligned}$$

where in line 2 we have used Binomial approximation $(1 + x)^n \approx 1 + nx$ when $x \ll 1$. Thus, the resulting motion is simple harmonic with angular frequency $\omega = \sqrt{\frac{3Mg}{R(M+m)}}$.

6 Equations of Constraint

Generally, when solving problems, you will find that the equations derived from Newton's laws alone are not sufficient to solve the problem. The main reason for this is that there are usually conditions imposed on the motion of the objects. To completely solve the system, these conditions must be written as equations alongside Newton's laws. The equations that describe these conditions on the object's motion are what we call equations of constraint. The simplest example of this would be two masses suspended by a rope on either side of a pulley. Here, alongside Newton's second law, we use another condition: the magnitudes of the accelerations of the two blocks are equal. This condition arises from the constraint that the rope is inextensible.



Observe the image above. If the radius of the pulley is R, then we can say the length of the rope is:

$$\begin{aligned} l &= (y_1 - y_3) + (y_2 - y_3) + \pi R \\ &= y_1 + y_2 - 2y_3 + \pi R \end{aligned}$$

Differentiating this equation twice with respect to time yields:

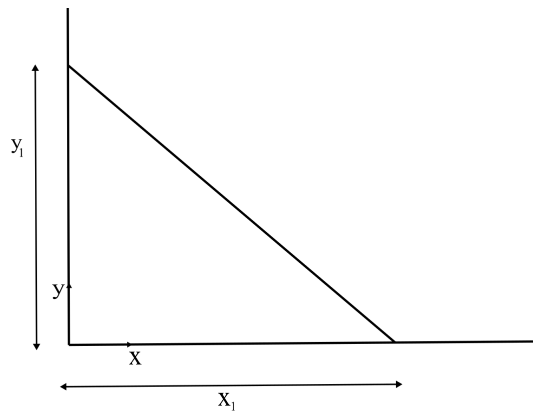
$$\ddot{l} = \ddot{y}_1 + \ddot{y}_2 - 2\ddot{y}_3 + 0$$

Now, our given condition is that the rope is inextensible. Furthermore, y_3 will not change with time, since the distance from the ceiling to the pulley is fixed. Because of this, $\ddot{y}_3 = 0$ and $\ddot{l} = 0$. Therefore,

$$\ddot{y}_1 = -\ddot{y}_2$$

Notice that $\ddot{y}_1$ and $\ddot{y}_2$ are the accelerations of block 1 and block 2, respectively. Since we defined the downward direction as positive, the negative sign essentially indicates that if one block moves down, the other will move up, and the magnitude of their accelerations will be identical.

When a constraint involves an inextensible rope or something whose value does not change over time, a very effective method for finding the constraint equation is to first express that unchanging quantity in terms of other variables, and then take its derivative with respect to time. You need to differentiate once if you require a relationship between the velocities of the objects, and twice if you need a relationship between their accelerations. As another example, consider a rigid rod leaning against a wall.



We need to find the relationship between the velocities of the two ends of the rod. Here again, regarding the length of the rod, we can write:

$$l^2 = x_1^2 + y_1^2$$

Differentiating with respect to time,

$$2l\dot{l} = 2x_1\dot{x}_1 + 2y_1\dot{y}_1$$

Substituting $\dot{l} = 0$ and eliminating the constants,

$$\frac{\dot{x}_1}{\dot{y}_1} = -\frac{y_1}{x_1}$$

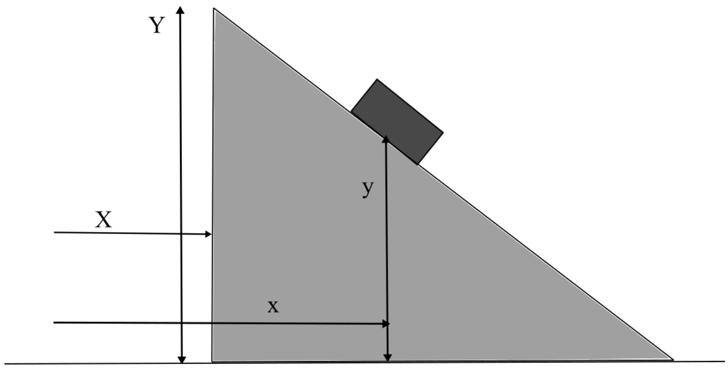
In the kinematics chapter, we saw that the components of velocity along the rod at both ends of a rigid body are equal. You can manipulate the equation we just found a bit further to show that it is essentially stating the exact same thing: the velocity components along the rod are equal at both ends.

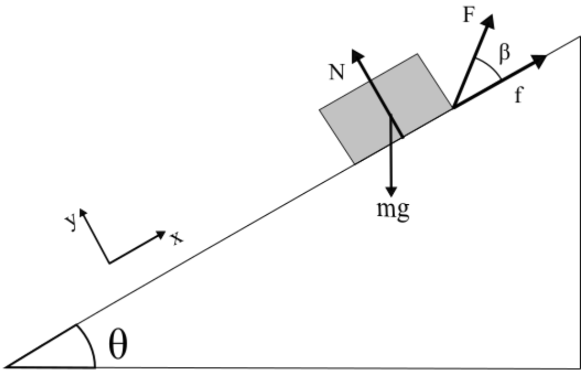
Using differentiation to find the equation of constraint in this manner always works, but it is not always the best method. There is another method where the system is analyzed by slightly displacing one of the objects. This is highly effective for pulleys. Think about our previous pulley; if you raise one block by 1 meter, the other block will descend by 1 meter. So, if you move one block down by x meters per second, the other will rise by x meters per second. You can establish a relationship between accelerations in the same way. While this method is generally fun, there is a higher chance of making errors using this rule in slightly more complicated systems; therefore, it is more advantageous to use the previous method.

Apart from these, there is another type of equation of constraint that appears occasionally; that is the condition for an object to remain on a specific surface or axis. For example, consider a train. The train is constrained to move only along the railway tracks. In this case, the easiest approach is to define the x-axis along the railway track (even if it is curved, it is not an issue), and then simply apply Newton's second law along this x-axis. As another example, consider an object on a fixed, inclined plane. Here too, the easiest way to handle the constraint is to align the x-axis parallel to the plane and the y-axis perpendicular to it. Then, the equation of constraint will be $y = \dot{y} = \ddot{y} = 0$! However, this approach does not work in all scenarios. Think about the previous inclined plane example, but this time the inclined plane itself is moving with a non-zero acceleration. In that case, we could still align the x-axis parallel to the inclined plane, but that frame would be non-inertial. It is possible to work in this frame, but we generally try to avoid doing so. In such a situation, the condition we can use is that the angle θ the inclined plane makes with the horizontal remains constant. From the figure below, we can write:

$$\tan\theta = \frac{Y-y}{x-X}$$
$$(x-X)\tan\theta = Y-y$$
$$(\ddot{x}-\ddot{X})\tan\theta = \ddot{Y}-\ddot{y}$$

where $(\ddot{x},\ddot{y})$ is the acceleration of the block in the lab frame, and $(\ddot{X},\ddot{Y})$ is the acceleration of the inclined plane in the lab frame. Note that these equations never depend on the origin of the coordinates, only on the orientation of the axes and the reference frame. Now, let us finish this chapter by pulling everything together with a few examples.





Example: An object of mass m is placed on a fixed plane inclined at an angle θ. The coefficient of static friction between the plane and the object is μ_s, and $\mu_s < \tan\theta$. What is the minimum force, and in which direction must it be applied, to keep the object stationary on the plane?
Solution: First, let us draw a force diagram: In the figure, force F is being applied at an angle β relative to the plane; force f is the static friction force. You might have many questions here regarding the direction of the friction. Think about it: since $\tan\theta > \mu_s$, the friction force alone cannot keep the object stationary. Consequently, the object has a tendency to slide downwards. At this point, we want to apply the smallest possible force F so that the block no longer slides down. Since the block has a tendency to slide downwards, even with the application of this minimal force, the tendency will remain to slide downwards; therefore, the friction will act upwards. For this reason, one should always think carefully before determining the direction of static friction. If you still have doubts about the direction of friction, proceed with your calculations using a $\pm$ sign in front of the friction force.

Writing the forces along the x-axis:

$$\sum F_x = f + F\cos\beta - mg\sin\theta = 0$$

And along the y-axis:

$$\sum F_y = N + F\sin\beta - mg\cos\theta = 0$$

Since we are looking for the condition where the object is on the verge of sliding, the friction force here will be at its maximum, meaning $f = \mu_s N$. Solving the previous equations together for F yields:

$$F = \frac{mg(\sin\theta - \mu_s\cos\theta)}{\cos\beta - \mu_s\sin\beta}$$

We are looking for the minimum value of F for various values of β. Since the numerator of the fraction does not depend on β, the value of F will be minimal when $(\cos\beta - \mu_s\sin\beta)$ is maximal. Differentiating with respect to β, you will find its minimum value occurs when $\tan\beta = -\mu_s$. Therefore,

$$F_{min} = \frac{mg(\sin\theta - \mu_s\cos\theta)}{\frac{1}{\sqrt{1+\mu_s^2}} - \frac{\mu_s\cdot(-\mu_s)}{\sqrt{1+\mu_s^2}}} = \frac{mg(\sin\theta - \mu_s\cos\theta)}{\sqrt{1+\mu_s^2}}$$

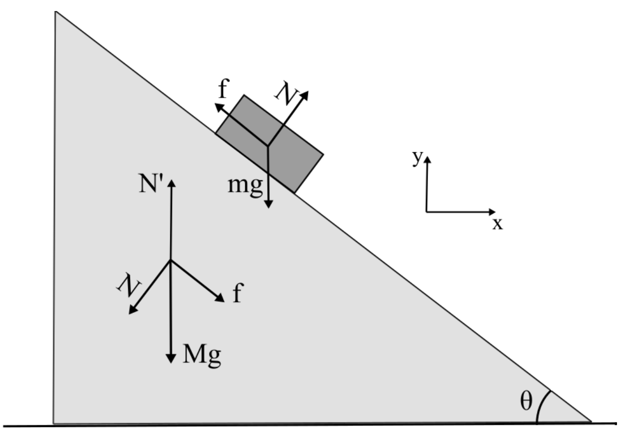
If you were unsure about the direction of friction and calculated using $\pm$, your F should come out to be:

$$F = \frac{mg(\sin\theta \mp \mu_s\cos\theta)}{\cos\beta \mp \mu_s\sin\beta}$$

From this, you can calculate F separately for both cases and show that between the two minimum values, the force is minimized when the direction of friction is exactly as we initially assumed it to be.

Example: There is a plane of mass M, inclined at an angle θ, on a frictionless horizontal surface. A block of mass m is placed on top of the plane. If the block of mass m is released, what will be the acceleration of the inclined plane? The coefficient of kinetic friction between the block and the plane is μ, where $\mu < \tan\theta$.

Solution: Once again, we will first draw a force diagram. The forces acting will be (i) the normal force N between the block and the plane, (ii) the friction f between the block and the plane, (iii) the weights of the block and the plane, and (iv) the normal force N' between the inclined plane and the horizontal surface. What you must be careful about in this problem is that the reaction forces to the forces acting on the block will act on the plane. You will understand this by looking at the force diagram below.



Note that since $\mu < \tan\theta$, the block will be in motion downwards, so the kinetic friction will act upwards. Furthermore, since it is kinetic friction, we can say $f = \mu N$. Let us first write the forces acting on the object along the x and y axes:

$$m\ddot{x} = N\sin\theta - \mu N\cos\theta$$

$$m\ddot{y} = N\cos\theta + \mu N\sin\theta - mg$$

Applying Newton's second law to the inclined plane along the x-axis:

$$M\ddot{X} = \mu N\cos\theta - N\sin\theta$$

We will not apply Newton's second law along the Y-axis because while it would give us a new equation, it would introduce a new variable N'. Consequently, using this equation would not benefit us. We currently have three equations, but four unknown variables ($\ddot{x}, \ddot{y}, \ddot{X}$ and N). Another equation comes from the constraint that the block will remain on the inclined plane. We previously derived such an equation and found:

$$(\ddot{x} - \ddot{X})\tan\theta = \ddot{Y} - \ddot{y}$$

where $\ddot{x}, \ddot{y}$ denote the acceleration of the block and $\ddot{X}, \ddot{Y}$ denote the acceleration of the inclined plane. In this case, the acceleration of our inclined plane along the y-axis is $\ddot{Y} = 0$. Therefore,

$$(\ddot{x} - \ddot{X})\tan\theta = -\ddot{y}$$

We now have four equations. The simplest way to solve them would be to substitute $\ddot{x}, \ddot{y}, \ddot{X}$ from the previous equations into this final equation. Then, after quite a bit of calculation, you will get:

$$N = \frac{Mmg\cos\theta}{M + m\sin^2\theta - \mu m\sin\theta\cos\theta}$$

Now we can determine the values of the remaining variables by substituting N into any of the previous three equations. Upon calculating, you will get:

$$\ddot{X} = \frac{mg\cos\theta(\mu\cos\theta - \sin\theta)}{M + m\sin^2\theta - \mu m\sin\theta\cos\theta}$$

You can substitute $\mu = \tan\theta$ into both N and $\ddot{X}$ to see that the familiar results $N = mg\cos\theta$ and $\ddot{X} = 0$ are obtained (since in this case, the block is stationary).

Notice that we did not set the x-axis parallel to the surface here. The reason is that there will be acceleration along this direction now, so there is no special advantage to using it.

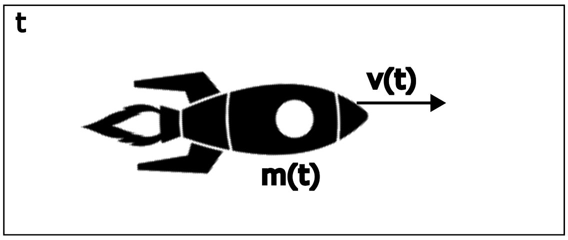
7 Variable Mass Systems

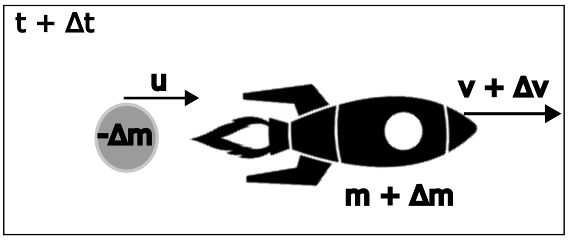
So far in our discussion, we have defined our systems by their mass. We can always choose our system in such a way that its total mass remains constant. However, without discussing how to apply Newton's laws to variable mass systems, these notes would remain incomplete. For example, let us begin our discussion with the familiar case of a rocket. Suppose there is a rocket in space whose mass at time t is $m(t)$ and whose velocity is $\vec{v}(t)$. The rocket ejects a portion of its mass (fuel) $-\Delta m$[5] at a velocity $\vec{v}_{\text{rel}} = \vec{u} - \vec{v}$ relative to itself, where $\vec{u}$ is the

[5] Since fuel is ejected from the rocket, the change in mass is $\Delta m < 0$, which means $-\Delta m > 0$.

velocity of the fuel and $\vec{v}$ is the velocity of the rocket. We are interested in determining the motion of the remaining mass of the rocket; we are not concerned with the motion of the exhaust fuel. In this case, if we try to apply Newton's second law strictly to the remaining portion of the rocket, the mass of our system is changing! If we instead consider the system to include the fuel, we can apply Newton's second law in its standard form. In that case, the initial momentum $\vec{P}_i$ and the final momentum $\vec{P}_f$ are respectively:

$$\vec{P}_i = m(t)\vec{v}(t) \qquad \text{and} \qquad \vec{P}_f = (m + \Delta m)(\vec{v} + \Delta\vec{v}) - \vec{u}\Delta m$$





Applying Newton's law to the entire system, we can say:

$$\begin{aligned}
\vec{F}_{\text{ext}} &= \lim_{\Delta t \to 0} \frac{\Delta\vec{P}}{\Delta t} = \lim_{\Delta t \to 0} \frac{\vec{P}_f - \vec{P}_i}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{\vec{v}\Delta m + m\Delta\vec{v} - \vec{u}\Delta m}{\Delta t} \\
&= m\frac{d\vec{v}}{dt} + (\vec{v} - \vec{u})\frac{dm}{dt} \\
&= m\frac{d\vec{v}}{dt} - \vec{v}_{\text{rel}}\frac{dm}{dt}
\end{aligned}$$

A few points regarding this equation need to be clarified. First, $\frac{dm}{dt}$ is the rate of change of the rocket's mass. That is, if the rocket's mass decreases, $\frac{dm}{dt} < 0$. Additionally, $\vec{v}_{\text{rel}}$ is the velocity of the lost (or gained) mass relative to our system.
If you made a seemingly minor mathematical mistake, you might have obtained the following equation:

$$\vec{F}_{\text{ext}} = \frac{d\vec{p}}{dt} = \frac{d(m\vec{v})}{dt} = m\frac{d\vec{v}}{dt} + \vec{v}\frac{dm}{dt} \qquad \text{(Incorrect)}$$

This equation cannot possibly be the correct equation for force because it violates Galilean invariance! In Newtonian mechanics, the magnitude of force must remain the same in any inertial reference frame. However, because $\vec{v}$ is present in this force equation, the calculated force would differ in different reference frames. Therefore, this equation can in no way be

correct.[6] But, what exactly was our mistake here? We merely applied a rule of calculus! I will leave it up to you to try and find the answer to this question!
The derived equation above is correct, but by now you have probably realized that applying it requires paying attention to a few subtle details. Therefore, in practice, when dealing with the motion of an object with changing mass, it is often better to proceed by applying Newton's second law to a larger, closed system rather than using our derived equation directly. Directly applying the formula might save you some time, but solving the problem by looking at the change in the momentum of the entire system will help you build much better intuition and a deeper understanding of both the problem itself and Newton's laws.
Notice that when $\frac{dm}{dt} = 0$, the mass of the object is not changing. Furthermore, when $\vec{v}_{\text{rel}} = 0$, the separated mass is moving at the exact same velocity as the system. In both cases, we expect to recover our familiar $F = ma$, and indeed our equation perfectly reduces to that. However, this second condition does not hold true for the equation obtained through simple differentiation.

Example: An ideal uniform rope of mass M and length L is held vertically over a weighing machine such that its lowest point is just barely touching the scale. If the upper end of the rope is released, the rope begins to fall onto the machine. At the exact moment when a length x of the rope is resting on the scale, what is the reading on the weighing machine?
Solution: To solve this problem, we must first understand that the reading of a weighing machine is equal to the downward vertical normal force (N) exerted by the object on the scale. Since the total length of the rope is constant, every point on the falling portion must have the exact same velocity, and we further assume that as soon as any part of the rope hits the machine, it instantly comes to rest. Immediately after the highest point of the rope is released, the tension at that point becomes 0; as a result, this point will fall with an acceleration of g purely under the influence of gravity. Therefore, when a length x of the rope is resting on the machine, the velocity of the remaining falling portion is $\sqrt{2gx}$.
To apply Newton's law, let us take the upward direction as positive. In this case, the net external force on the portion of the rope currently on the weighing machine is $(N - \lambda x g)$, where $\lambda = \frac{dm}{dx}$ is the linear mass density of the rope. Furthermore, even though mass is continually being added to the scale, this resting portion of the rope remains stationary, so its acceleration is zero. The relative velocity of the falling portion with respect to the weighing machine is $v_{\text{rel}} = -\sqrt{2gx}$. Thus, we can state:

$$\vec{F}_{\text{ext}} = m\frac{d\vec{v}}{dt} - \vec{v}_{\text{rel}}\frac{dm}{dt}$$

$$N - \lambda x g = 0 + \sqrt{2gx}\frac{dm}{dx}\frac{dx}{dt} = \sqrt{2gx}\lambda\sqrt{2gx}$$

$$N = 3\lambda g x = 3Mg\frac{x}{L}$$

The reading on the weighing machine reaches its maximum when the entire rope has fallen onto the scale ($x = L$), at which point the reading is $3Mg$. Now, your task is to try solving this exact same problem without explicitly using our extended Newton's formula. Although the solution will be slightly longer, it will be significantly more insightful.

[6] Notice that our correct equation contains $\vec{v}_{\text{rel}}$, not $\vec{v}$. Although the magnitude of velocity varies across different reference frames, the relative velocity remains identical in all frames. Thus, our correct equation does not suffer from this issue.

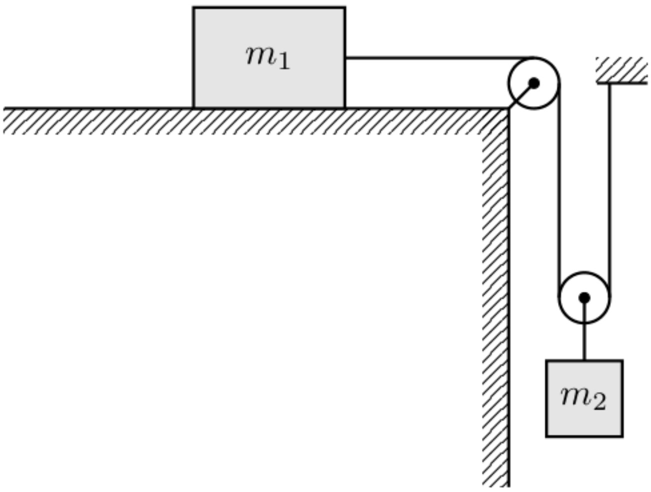
8 Exercise

8.1 Particle falling off a hemisphere

A small block of mass m is placed at the highest point of a fixed, frictionless hemisphere of radius R. The block is given a very small push. At what angle with the vertical will the block lose contact with the surface of the hemisphere? What is the speed of the block at the moment it loses contact?

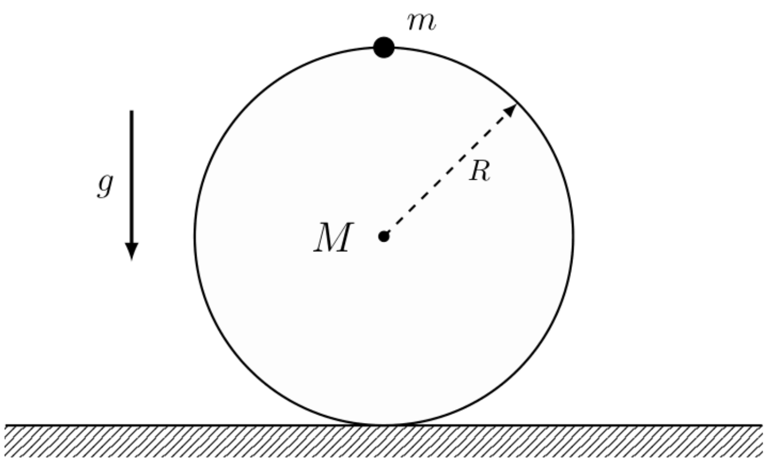
8.2 Pulley and Constrains

Determine the acceleration of both blocks and the tension in the string in the figure below.



8.3 More on constrains

A point mass m is attached to the highest point of a ring of mass M and radius R. The ring can move without friction on a horizontal plane (left-right in the figure below), and the point mass is fixed to a single point on the circumference of the ring. When the line connecting the point mass and the center of the ring makes an angle θ with the vertical, what is the horizontal acceleration of the ring? What is the normal force exerted by the ground on the ring? Can the ring lift off the ground for any value of M?



8.4 Water Stream

Water ejects horizontally from a fire engine's hosepipe at a speed v and perpendicularly strikes a vertical wall. The cross-sectional area of the pipe is A and the density of water is ρ. What is the force exerted by the water on the wall if:

1. The water flows down the wall after striking it.
2. The water flows down the wall after striking it, and the wall is moving towards the pipe at a speed u.

8.5 Flying over a bridge

A bridge forms a part of a circle of radius R meters. What is the maximum speed at which a car can travel over this bridge so that it never loses contact with the road? If the bridge were a part of the parabola $y = -ax^2$ from $x = -l$ to $x = l$ instead of a circle, what would be the maximum speed to cross the apex of the bridge without losing contact?